

Example 8: NR Restaurant, Single Zone, DOAS Units Configured as Rooftop Unit

SYSTEM TAG: T24ACM-SZVAVAC

PROJECT VISION

California Title 24 compliance modeling requires translating mechanical system designs into CBECC 2025 software objects, a process for which little practical guidance exists. Developed as part of the *Addressing Complex HVAC Systems in Code Compliance* project, funded by Southern California Edison through the CalBEM working group, each example walks through how a recognizable HVAC system type translates into CBECC objects, what Title 24 2025 Part 6 requires, and what assumptions were made.

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Example 8: NR Restaurant, Single Zone, DOAS Units Configured as Rooftop Unit

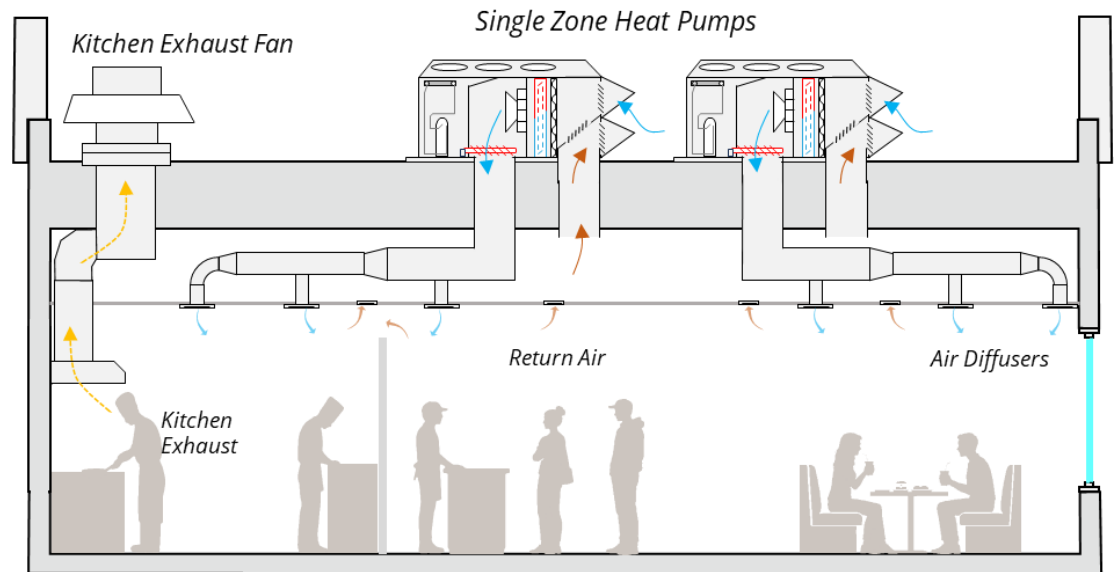
System Tag: T24ACM-SZVAVAC

Title 24 2025 ACM: System 7a

Mechanical Description

The building is conditioned by two independently controlled packaged rooftop units, each serving a single occupied space; one the dining area, one the food preparation area. Both units are configured as mixed-air, rooftop units with AC and furnaces. A kitchen exhaust system maintains building pressure, relieving air through the hoods. The kitchen hood operates as a constant-volume commercial kitchen exhaust fan, ducted directly to outdoors. No heat recovery is provided on the exhaust air.

The primary rooftop equipment selected in this example can be configured as mixed-air rooftop units or as dedicated outdoor air systems, and was labeled by the manufacturer as 'DOAS units'. This example demonstrates how the configuration of a system should always reflect the architecture of the airflow and control, rather than the label of products specified by a manufacturer.



The diagram shows a standard fast-food restaurant, with one single zone unit serving the dining area, another serving the serving and cooking area. The exhaust fan runs to maintain indoor air quality with air transferring from the dining area as needed to maintain exhaust.

Mechanical Details

Two SZ VAV AC systems were applied to the fast-food restaurant prototype, with both thermal zones conditioned by their respective packaged rooftop unit. The Kitchen zone is also served by a continuous hood exhaust.

Ventilation: Outdoor air delivered through each RTU as a fraction of the total supply air to each zone.

Ventilation control: The Dining zone uses CO₂ based demand-controlled ventilation with a VAV terminal modulating supply between min and max airflow. The Kitchen zone ventilation is constant with VAV terminal modulating supply between min and max.

Cooling: Single-stage DX cooling on Dining RTU and 2-stage DX cooling on the Kitchen RTU.

Heating: Natural gas furnace at each rooftop unit, mixed fuel equipment matching real-world examples for this occupancy type.

Capacity control: VSD supply fan at each unit modulated to meet zone load through VAV no-reheat terminal box.

Economizing: Yes. Differential enthalpy economizer on both rooftop units, integrated with mechanical cooling.

Heat Recovery: None. Kitchen hood exhaust is exempt from heat recovery requirements.

Mechanical Capacities and Sizing

Systems were sized based on the as-built equipment selection documented in the source building's actual mechanical drawings given the proximity of the building size to the prototype for both zones. Outdoor air for the Dining zone was sized to the zone ventilation requirements, with CO₂-based DCV at the VAV terminal modulating supply between a minimum and maximum airflow. Outdoor air for the Kitchen zone was set to match both the zone ventilation requirement for kitchen/cooking occupancy and the makeup air requirement against the continuous hood exhaust. The Kitchen VAV terminal holds a constant minimum airflow floor with no CO₂ DCV applied. Cooling efficiencies were entered at generic values for each unit; both gas furnaces were entered with mandatory minimum AFUE matching drawing specs.

Kitchen Exhaust System

Exhaust (cfm)	Control	Hood Type	Air Path
2,400	Constant Volume	Wall Canopy, Heavy, 10 ft	Ducted to outdoors not in return air path

Zone Airflow Balance

Zone	Sup. (cfm)	Min OA (cfm)	Return (cfm)	Exh. (cfm)	Net Pressurization
Dining	2,000	1,200	500 to 1300	—	Positive (700 cfm transfer air to Kitchen)
Kitchen	4,500	1,700	Up to 2,800	2,400	Balanced (pulling 700 cfm air from Dining)
Building Total	6,500	2,900		2,400	Net +500 cfm OA

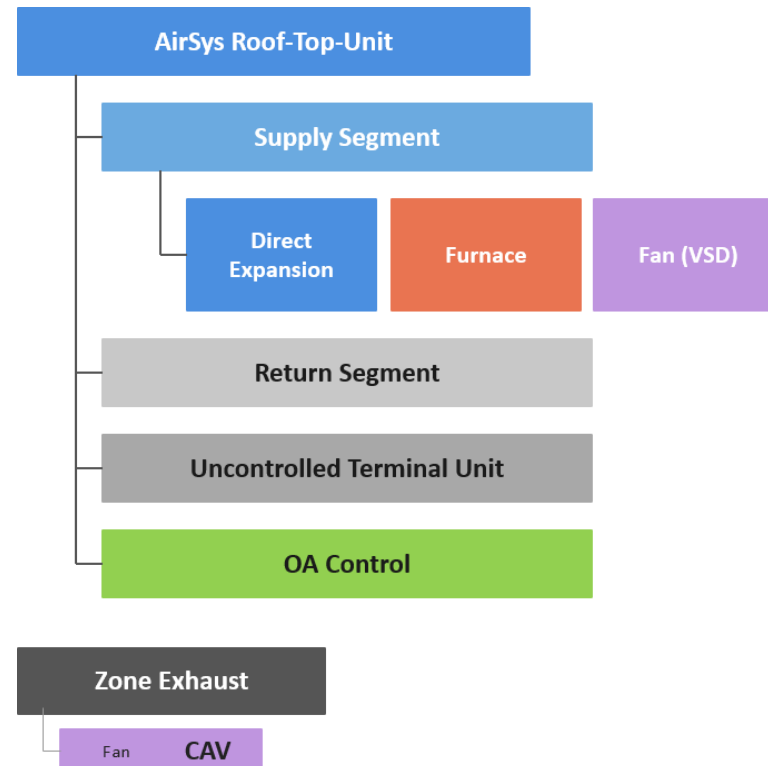
Code Compliance Software Configuration

Each RTU was entered as a Package SZ VAV AC (SZVAVAC) air system serving a single thermal zone, with return air ductwork explicitly modeled (return air segment Path = Direct on each system). Outdoor air control was configured at the OA controller object for each system: DifferentialEnthalpy economizer, Integrated, with 71°F dry-bulb and 28 Btu/lb enthalpy high-limit lockouts. The Dining OA controller carries MaxOARat: 1.0 to allow full economizer modulation; the Kitchen OA controller has no MaxOARat set, fixing OA volume at the terminal minimum.

Generic efficiency values were entered for the DX cooling coils, IEER 14.6 on RTU-1 serving the dining area and 14.0 on RTU-2 serving the kitchen, rather than the manufacturer's published ratings. Those published ratings are established under AHRI 920 dedicated outdoor air conditions, while the compliance minimums in Table 140.4-A are set on the AHRI 340/360 basis for the system types these units were modeled as. Resolving that rating-basis mismatch is outside the scope of this project. The model therefore represents the system architecture rather than the performance of the specific equipment selected.

The Kitchen thermal zone was assigned the KitchenExhaust system reference (ExhSysRef = KitchenExhaust) and configured with FullyDuctedExhFlag = 0 on that exhaust system,

instructing the simulation engine to permit transfer air from adjacent zones rather than treating the hood as a closed return loop. Hood inputs at the Kitchen space include KitExhHoodFlow (2,400 cfm), KitExhHoodLen (10 ft), KitExhHoodStyle (WallMountedCanopy), and KitExhHoodDuty (Heavy).



CBECC Object Locations

The model is made up of three systems: two air systems, one for dining, one for the kitchen, and a zone system for the kitchen exhaust.

Air Systems

The dining system is shown here for key inputs on the air-system input screen.

The unit was assumed to be ducted on the roof in an unconditioned space with duct insulation and duct leakage testing.

Air System Data	Ducts	Acceptance Certificates	Fan Power Allowances	Sizing
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Currently Active Air System: Dining - RTU / DOAS 1 ☐ Acceptance Test Required?

Name: Dining - RTU / DOAS 1

Type: SZVAVAC Status: New

Sub Type: Packaged3Phase Count: 1

Control Zone: Dining Thermal Zone

Description:

Fan Control: Continuous Night Cycling: CycleOnCallPrimaryZone

Design OA Flow: 619 cfm ☐ 100% OA System

Control Type: DDCToZone

☐ Optimum Start
☐ Complex Mechanical System?
(per Section 10-102)

Cooling Heating Supply Flow

Design Supply Air Temp: 55.0 °F 85.0 °F 2,000 cfm

Net Capacity*: 98,700 Btu/h 95,423 Btu/h

*Reflects capacity of a single system if Count is >1. Heating capacity includes terminal units.

Fan Position: DrawThrough

Supply Temp Control: NoSATControl

Cooling Heating

Fixed Supply Temp: °F °F

Setpoint Temp Sch.: - none - - none -

Reset Supply High: °F @ Outdoor Temp: °F Reset Supply High: °F @ Outdoor Temp: °F

Reset Supply Low: °F @ Outdoor Temp: °F Reset Supply Low: °F @ Outdoor Temp: °F

--- Sizing Parameters ---

Air System Data	Ducts	Acceptance Certificates	Fan Power Allowances	Sizing
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Currently Active Air System: Kitchen-RTU / DOAS 2

Name: Kitchen-RTU / DOAS 2

Type: SZVAVAC

☒ Duct leakage and sealing required per 140.4(l) ☒ Duct leakage will be field verified per NA1 and NA2

☐ Duct is Existing

Duct Location: Unconditioned

Duct Insulation: 8.0 °F-ft2-h/Btu Code Minimum: 8.0

Duct insulation used for reporting and mandatory checks only.

Air System Fan Component

The fan airflow for each RTU was configured as a variable speed drive fan based on the design.

Zone	Supply Fan (cfm)	Outdoor Air (cfm)	Fan Eff. (W/cfm)	Economizer
Dining	2,000	1,200	0.746	Diff. Enthalpy
Kitchen	4,500	1,700	0.820	Diff. Enthalpy

Outdoor Air Controller

Each unit includes an economizer controlled based on enthalpy as shown below.

Fan Data

Currently Active Fan: **Dining Fan**

Name: **Dining Fan**

Control Method: **VariableSpeedDrive**

Classification: **Centrifugal**

Status: **New** Component Qty: **1**

Centrifugal Type: **AirFoil**

Capacity and Power (For single system/terminal if Component Qty > 1)

Flow Capacity: **2,000** cfm

Flow Minimum: **1,200** cfm

Position: **DrawThrough**

Modeling Method: **PowerPerUnitFlow**

Total Static Pressure: **3.570** inH2O

Fan Efficiency: **0.650**

Motor Brake HP: **1.731** hp **0.865** hp/1000cfm

Power Per Flow: **0.746** W/cfm

Total Power (Ref): **1.492** kW

Motor Information

Position: **InAirStream**

Nameplate HP: **2.000** hp

Type: **Open**

Pole Count: **4**

Motor Efficiency: **0.865**

Total Efficiency: **0.562**

Fan Energy Index (FEI): **1.000**

FEI Exception: **Not meeting any EXCEPTIONs**

Currently Active OA Control: **Kitchen OAControl**

Name: **Kitchen OAControl**

Status: **New** Component Qty: **1**

Supply AirSeg Ref: **Kitchen SupAirSeg** Heat Recovery Ref: **- none -**

Return AirSeg Ref: **Kitchen ReturnAirSeg**

Outside Air (For single system if Component Qty > 1)

Design OA Flow: **1,700** cfm Schedule Method: **FollowHVACAvailability** Delay OA Durin

Max OA Ratio: **1.00** Min Fraction Sch: **- none -**

Economizer Controls

Control Method: **DifferentialEnthalpy**

Integration: **Integrated**

High DB Temp Lockout: **71.0** °F

Low DB Temp Lockout: °F

High Enthalpy Lockout: **28.0** Btu/lb

Zone System, Kitchen Exhaust

Zone kitchen exhaust was configured with a zone system set to CommercialKitchen as the fan type.

The control type was assumed to be a constant flow control with a constant speed fan.

Zone System Data		Ducts	Acceptance Certificates	Fan Power Allowances		
Currently Active Zone System:				KitchenExhaust	☐ Acceptance Test Required?	
Name:	KitchenExhaust					
Type:	Exhaust	Status:	New	System Operation:	DecoupledFromSystem	
Exh. Type:	CommercialKitchen	Count:	1	Flow Schedule:	- none -	
Exh. Control:	ConstantFlowConstantSpeedFan					
Description:						☐ Complex Mechanical System? (per Section 10-102)
				Exh. System Flow:	2,400 cfm	
				Exh. System Power:	1.560 kW	
					0.650 ***cfm	

Thermal Zone Exhaust Control

The thermal zone component for the kitchen sets the link to the kitchen exhaust fan, as well as defines the source of makeup air. In this configuration the makeup air is taken from the adjacent zones in the building, the dining area.

Thermal Zone Data	Ventilation and Exhaust	Temp Control and Sizing	Daylighting																								
Currently Active Zone: Kitchen Thermal Zone																											
Ventilation (Outdoor) Air		Exhaust Air																									
Ventilation Standard: T24-2025		Exhaust System**: KitchenExhaust																									
Specification Method: Maximum		Control Type: VariableFlowConstantSpeedFan																									
Control Method: Fixed ☐ System Has DDC To Zone		Exhaust Source***: TransferFromAdjacentZones																									
Floor Area/Volume: 1,250.5 ft2 12,507.9 ft3		Exhaust Heat Recovery: - none -																									
Occupancy: 3.8 people 3.0 people/1000ft2																											
<table><thead><tr><th></th><th>Design</th><th>Code Min.</th></tr></thead><tbody><tr><td>Total Ventilation (cfm):</td><td>1,700.0</td><td>1,700.0</td></tr><tr><td>(cfm/ft2)*:</td><td>1.359</td><td>1.359</td></tr><tr><td>(ACH)*:</td><td>8.155</td><td>8.155</td></tr></tbody></table>			Design	Code Min.	Total Ventilation (cfm):	1,700.0	1,700.0	(cfm/ft2)*:	1.359	1.359	(ACH)*:	8.155	8.155	<table><thead><tr><th></th><th>Design</th><th>Code Min.</th></tr></thead><tbody><tr><td>Total Exhaust (cfm):</td><td>2,400.0</td><td>2,400.0</td></tr><tr><td>(cfm/ft2)*:</td><td>1.919</td><td>1.919</td></tr><tr><td>(ACH)*:</td><td>11.513</td><td>11.513</td></tr></tbody></table>			Design	Code Min.	Total Exhaust (cfm):	2,400.0	2,400.0	(cfm/ft2)*:	1.919	1.919	(ACH)*:	11.513	11.513
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*Flow rate shown is the total as a normalized value, not component values. See HELP for details.

**Do not model intermittent exhaust systems

*** 700.0 CFM ventilation air is transferred from adjacent zones

System Efficiency Tables

Category	Proposed Design	vs. Standard Baseline
System Type	Package SZVAVAC (both zones), VSD fan, partial-recirculation RTU configured with return air ductwork.	Standard assigns CV/SZ AC to Dining and SZ VAV AC to Kitchen; proposed uses VSD VAV on both zones.
Cooling	RTU-1 Dining: IEER 14.6, single-stage DX. RTU-2 Kitchen: IEER 14, two-stage DX.	Same DX equipment type per Table 140.4-A. Proposed entered at generic efficiencies.
Heating	Natural-gas furnace, AFUE 81%, at each RTU. No supplemental heat.	Same fuel and AFUE. NRCC C2 shows proposed losses 3.76 LSC on space heating vs. standard.
Supply Fan	VSD centrifugal: 0.75 W/cfm (Dining RTU), 0.82 W/cfm (Kitchen RTU).	Standard Dining: Constant volume fan. Kitchen: VSD. Proposed VSD on Dining saves 8.23 LSC on indoor fans.
VAV Terminal	VAVNoReheatBox at both zones. Dining modulates 1,200–2,000 cfm under CO2 DCV. Kitchen fixed at 1,700 cfm.	Standard Dining: ConstantVolume, no DCV. Kitchen terminal type same in both designs.
Economizer	Differential enthalpy, integrated with mechanical cooling, on both RTUs.	Standard uses FixedDryBulb 71°F. Differential enthalpy performs better on cool, humid CZ9 coastal mornings.
Photovoltaics	24.3 kWdc rooftop PV array.	NRCC C2 shows proposed gains of 1.29 LSC vs. baseline.

Note: Heat recovery — neither RTU crosses the 70% outdoor air fraction threshold; kitchen hood exhaust (2,400 cfm Type I) is exempt from heat recovery requirements. Domestic hot water — 30-gallon natural-gas storage water heater, UEF 0.65; same equipment type and fuel in both proposed and standard designs.

Relevant Energy Code Citations

Required Demand Control Ventilation

401.2.1.2.3 [Section 120.1(d)3]

The Dining zone exceeds the occupancy density limit for cafeteria / fast-food dining, requiring CO₂-based modulating ventilation airflow. The VAV terminal on RTU-1 carries minimum and maximum airflow setpoints and is controlled by a zone CO₂ sensor. The Kitchen zone is not subject to DCV — kitchen ventilation rates are driven by the hood makeup-air requirement and held constant.

Fan Power Budget, Packaged VAV Units

401.3.3.3 [Section 140.4(a)3D]

Each Package SZ VAV AC supply fan must meet the prescriptive fan-power limit for VAV systems in its capacity range. The modeled supply fan power (0.75 W/cfm on RTU-1 and 0.82 W/cfm on RTU-2) is consistent with the prescriptive limit for packaged VAV systems at design airflow.

Air-Side Economizer, Packaged Units

401.3.7 [Section 140.4(e)]

Each rooftop unit exceeds the 33,000 Btu/hr cooling-capacity threshold that triggers the economizer requirement. Both units are configured with differential enthalpy economizers integrated with mechanical cooling, satisfying the prescriptive requirement in Climate Zone 9. Differential enthalpy was selected over fixed dry-bulb to improve free-cooling availability during the cool, humid morning hours typical of coastal Climate Zone 9 operation.

Exhaust Air Heat Recovery, Kitchen Hood Exemption

401.3.18 [Section 140.4(q)]

Outdoor air systems delivering 3,000 CFM or more at 70% or greater outdoor air fraction would typically require exhaust air heat recovery. Neither RTU exceeds the 70% outdoor air threshold (RTU-1 at 60%, RTU-2 at 38%), and the kitchen hood exhaust is independently exempt from heat recovery requirements as a Type I commercial kitchen hood exhaust stream. No heat recovery is required or provided.

Kitchen Hood Exhaust, Type I

Section 140.9(b)

The 2,400 cfm Wall-Mounted Canopy Type I hood is classified as heavy-duty. Hood airflow per linear foot meets the prescriptive maximum, and the hood is modeled with continuous occupied-hour operation. No demand-controlled kitchen ventilation is applied — the hood operates at constant volume.

Minimum DX Cooling and Furnace Efficiency

Table 401.3-A [Table 140.4-A]

The DX cooling coil and gas furnace heating coil at each rooftop unit must meet minimum EER, IEER, and AFUE thresholds for their capacity ranges. Both modeled units carry generic cooling efficiency entries, described in the modeling note above, and meet the 81% AFUE mandatory minimum on the gas furnace.

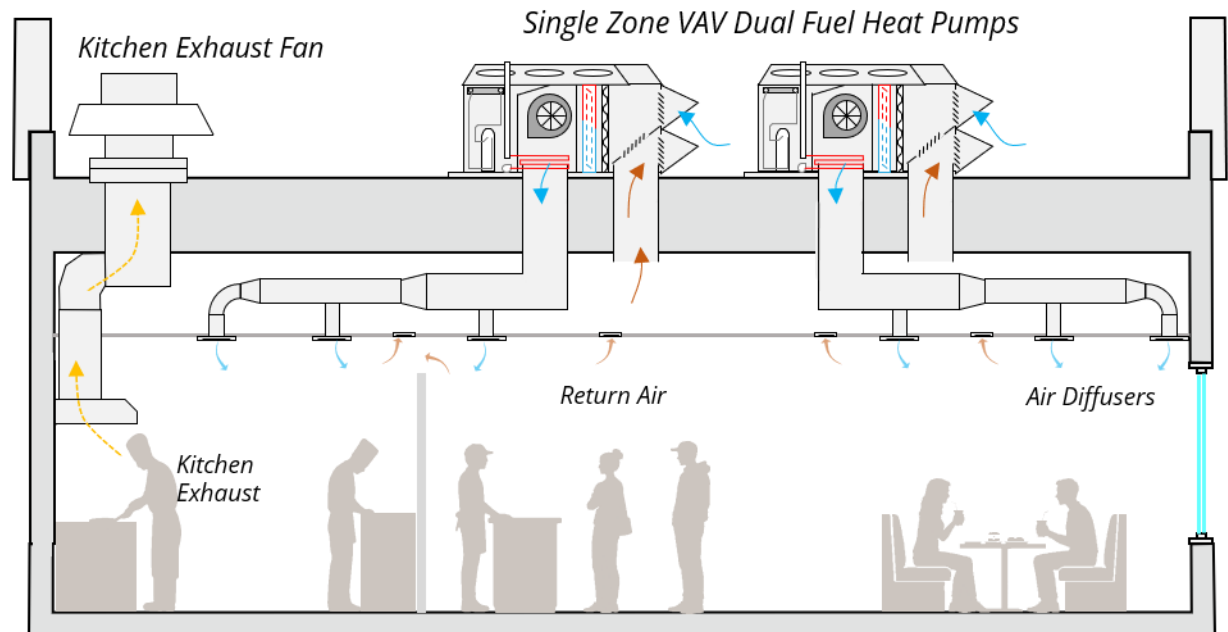
Alternate 8b: NR Restaurant, Single Zone Dual-Fuel Heat Pumps

System Tag: T24ACM-SZVAVDFHP

Title 24 2025 ACM: System 7c

Mechanical Description

This system is a single zone VAV airflow control dual-fuel heat pump configuration, with each of the two zones operating with their own unit. This configuration will consistently exceed the energy cost and source energy savings of an AC furnace configuration with the heat pump.



This example is one of eight developed as part of the *Addressing Complex HVAC Systems in Code Compliance* project. The full guide includes additional HVAC system types, alternative configurations, and working CBECC model files. All resources are available through the CalBEM working group and EnergyCodeAce platform.

